

Unit 7 Test Review

Roots, Factors, Grouping, & Sum/Difference of Cubes | TEKS: A2.7.D, A2.7.E

Reminders: • Zeros, roots, solutions, and x-intercepts all mean the same x-values where $f(x) = 0$. • If r is a zero, then $(x - r)$ is a factor. • The Factor Theorem says $f(a) = 0$ if and only if $(x - a)$ is a factor. • SOAP: **S**ame, **O**pposite, **A**lways **P**ositive.

Part A: Root–Factor Relationship

(Test Q1–Q2 style)

1. Let $f(x) = x^3 - 4x^2 - x + 4$. Which of the statements are true? **Select TWO.**

- A. $(x + 4)$ is a linear factor of f .
- B. $f(1) = 0$.
- C. $(1, 0)$ is an x-intercept of the graph of f .
- D. $x = -4$ is a zero of f .
- E. The remainder when $f(x)$ is divided by $(x - 4)$ is zero.

2. Consider the polynomial $x^3 + 2x^2 - 9x - 18$. Which of these are linear factors? Select the correct answer in each row.

	Linear factor	Not a linear factor
$(x + 2)$		
$(x - 3)$		
$(x + 3)$		
$(x - 9)$		

3. Use the given information to translate. (*No calculator needed.*)

- a. Zeros: $-6, 0, 4$ Factors: _____
- b. Factors: $(x - 7), (x + 1), (x - 3)$ x-intercepts: _____
- c. Roots: $-2, 5$ Factors: _____

4. The table for $f(x) = x^3 + 2x^2 - 5x - 6$ is shown. Determine the zeros, roots, and linear factors.

x	-4	-3	-2	-1	0	1	2	3	4
$f(x)$	-18	0	4	0	-6	-8	0	24	66

Zeros: _____ Roots: _____ Factors: _____

Part B: Factor by Grouping

(Test Q3, Q7 style)

5. Factor by grouping: $x^3 + 4x^2 - 9x - 36$

6. Factor by grouping: $2x^3 + 5x^2 - 8x - 20$

7. Factor by grouping: $6x^3 - 9x^2 + 10x - 15$

Part C: Sum & Difference of Cubes

(Test Q5–Q6 style)

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

SOAP: Same, Opposite, Always Positive

8. Factor: $x^3 - 216$

$a =$ _____ $b =$ _____

9. Factor: $27x^3 + 1$

$a =$ _____ $b =$ _____

10. Factor: $125x^3 - 8$

$a =$ _____ $b =$ _____

11. What are the factors of $x^3 + 512$?

The linear factor is _____ and the quadratic factor is _____.

Part D: Factor Completely

(Test Q4, Q8–Q9 style)

12. Kira and James are factoring $x^4 - 16$.

Kira says the only factors are $(x^2 + 4)$ and $(x^2 - 4)$.

James says the complete factorization is $(x^2 + 4)(x + 2)(x - 2)$.

Which student is correct, and why?

13. Factor completely: $x^3 + 6x^2 + x + 6$

14. It is known that two zeros of $f(x) = x^4 + 2x^3 - 13x^2 - 14x + 24$ are $x = 1$ and $x = -3$.

Which statements are true? **Select TWO.**

A. $(x - 1)$ is a factor of f .

B. $(x + 3)$ is a factor of f .

C. $(x^2 + 8)$ is a factor of f .

D. $(x^2 - 8)$ is a factor of f .

E. $(x - 3)$ is a factor of f .

Show your work — divide or factor to find the remaining quadratic:

15. Factor completely: $x^4 - 13x^2 + 36$

Hint: Treat x^2 as a single variable. What two numbers multiply to 36 and add to -13 ?

Part E: Quick Hits

(Mixed review — timed practice)

16. If $(x - 5)$ is a factor of $f(x)$, then $f(5) =$ _____

17. If $f(2) = 0$, then which is a factor of $f(x)$?

A. $(x + 2)$

B. $(x - 2)$

C. $(2x)$

D. $(x^2 - 2)$

18. The zeros of a polynomial are -3 , 0 , 2 , and 7 . How many linear factors does it have?

A. 2

B. 3

C. 4

D. 7

19. Which expression is a factor of $8x^3 + 27$?

A. $(2x - 3)$

B. $(2x + 3)$

C. $(4x^2 + 6x + 9)$

D. Both B and C

20. The graph of $f(x)$ crosses the x-axis at $x = -4$, $x = 1$, and $x = 6$. In factored form, $f(x)$ could be:

A. $(x + 4)(x - 1)(x - 6)$

B. $(x - 4)(x + 1)(x + 6)$

C. $(x + 4)(x + 1)(x + 6)$

D. $(x - 4)(x - 1)(x - 6)$

ANSWER KEY — Teacher Use Only**Part A: Root–Factor Relationship**

1. **B and E.** $f(x) = x^3 - 4x^2 - x + 4$. Factor by grouping: $x^2(x - 4) - 1(x - 4) = (x - 4)(x^2 - 1) = (x - 4)(x + 1)(x - 1)$. Zeros are 4, -1, 1. So $f(1) = 0$ ✓ and the remainder when dividing by $(x - 4)$ is zero ✓.

A is wrong: $(x + 4)$ is not a factor (zero would be -4). C: (1, 0) IS an x-intercept, so C is also true. *Note: B, C, and E are all correct. Students selecting any two of B/C/E should receive credit.*

2. Factor: $x^2(x + 2) - 9(x + 2) = (x + 2)(x^2 - 9) = (x + 2)(x + 3)(x - 3)$.

$(x + 2)$: **Linear factor.** $(x - 3)$: **Linear factor.** $(x + 3)$: **Linear factor.** $(x - 9)$: **Not a factor.**

- 3a. $(x + 6)$, x , $(x - 4)$ [sign flips: zero $-6 \rightarrow$ factor $(x + 6)$, etc.]

- 3b. $(7, 0)$, $(-1, 0)$, $(3, 0)$

- 3c. $(x + 2)$, $(x - 5)$

3. Zeros: -3, -1, 2 Roots: -3, -1, 2 Factors: $(x + 3)(x + 1)(x - 2)$

Part B: Factor by Grouping

5. $x^2(x + 4) - 9(x + 4) = (x + 4)(x^2 - 9) = (\mathbf{x + 4})(\mathbf{x + 3})(\mathbf{x - 3})$ *Don't forget to factor the difference of squares!*

6. $x^2(2x + 5) - 4(2x + 5) = (2x + 5)(x^2 - 4) = (\mathbf{2x + 5})(\mathbf{x + 2})(\mathbf{x - 2})$

7. $3x^2(2x - 3) + 5(2x - 3) = (\mathbf{2x - 3})(\mathbf{3x^2 + 5})$ $3x^2 + 5$ *doesn't factor further.*

Part C: Sum & Difference of Cubes

8. $x^3 - 216$: $a = x, b = 6$. $(\mathbf{x - 6})(\mathbf{x^2 + 6x + 36})$

9. $27x^3 + 1$: $a = 3x, b = 1$. $(\mathbf{3x + 1})(\mathbf{9x^2 - 3x + 1})$

10. $125x^3 - 8$: $a = 5x, b = 2$. $(\mathbf{5x - 2})(\mathbf{25x^2 + 10x + 4})$

11. $x^3 + 512$: $a = x, b = 8$ (since $8^3 = 512$). Linear: $(\mathbf{x + 8})$. Quadratic: $(\mathbf{x^2 - 8x + 64})$.

Part D: Factor Completely

12. **James is correct.** $x^4 - 16 = (x^2 + 4)(x^2 - 4)$, but $x^2 - 4$ is a difference of squares and factors further into $(x + 2)(x - 2)$. Kira didn't factor completely. Complete: $(x^2 + 4)(x + 2)(x - 2)$.

13. Group: $x^2(x + 6) + 1(x + 6) = (\mathbf{x + 6})(\mathbf{x^2 + 1})$ $x^2 + 1$ *doesn't factor over reals.*

14. **A and B.** Since $x = 1$ is a zero, $(x - 1)$ is a factor ✓. Since $x = -3$ is a zero, $(x + 3)$ is a factor ✓. Dividing: $x^4 + 2x^3 - 13x^2 - 14x + 24 \div (x - 1)(x + 3) = x^2 + 2x - 8 = (x + 4)(x - 2)$. So D is wrong ($x^2 - 8$ is not a factor). Full factorization: $(x - 1)(x + 3)(x + 4)(x - 2)$.

15. $x^4 - 13x^2 + 36 = (x^2 - 4)(x^2 - 9) = (\mathbf{x + 2})(\mathbf{x - 2})(\mathbf{x + 3})(\mathbf{x - 3})$

Part E: Quick Hits

16. **0** (Factor Theorem: if $(x - 5)$ is a factor, then $f(5) = 0$)

17. **B** ($f(2) = 0$ means $(x - 2)$ is a factor)

18. **C** (4 zeros = 4 linear factors)

19. **D** ($8x^3 + 27 = (2x)^3 + 3^3 = (2x + 3)(4x^2 - 6x + 9)$. Both B and C are factors.)

20. **A** (x-intercepts at -4, 1, 6 give factors $(x + 4)(x - 1)(x - 6)$)